



Hoverboards...Again

Hoverboard Technologies designs new prototype with a gyroscope which balances the rider on a single wheel. <http://dgit.in/hoverbrdwheel>



5G Testing by Verizon

The telecom operator is in the phase of testing 5G, but at a cost. Know more here: <http://dgit.in/Verizon5G>

Space Age

Although fusion is still theoretically possible current processes produce post fusion element plus a neutron. Since neutrons are neutral, you can't control where they end up. Let's just say that is an eventuality worse than Fukushima. Adding Tritium during fusion gives post fusion element plus a proton. Because a proton is positive it can be channelled and collected elsewhere using an electromagnetic field.

The energy generated from fusion process is staggering to say the least. The current estimates of the moon's reserves are estimated to meet all of the Earth's energy requirements for 1000 years at least. American scientists surmise that a shuttle full of Extracted and refined He-3 can power the all of USA for a year.

The Political Angle

In 1967, the Outer Space Treaty was signed, which stated that the exploration and use of outer space shall be carried out for the benefit of all countries and shall be the province of all mankind. Does this mean that the moon belongs to everyone? Or no one? Or are we going to duel it out? Is being debated by Space Lawyers. And Yes, there is such a vocation. There are only three major competitors in today's geopolitical theatre who are capable of participating in this space race. The West, Chinese and Arabic nations.

The West is spearheaded by the US and consists of the European nations. It has the support of the top corporations of the world working together, as well as competing with each other in various industries. Moon Express, SpaceX, Google Lunar XPrize, Lockheed Martin, Boeing among quite a few other companies built on top of the pioneer-

ing achievements accomplished by NASA.

In the West, even startups are attempting to land a small rover to the moon. They clearly do have the infrastructure, but their economy is giving way to other economies with cheaper labour. A paper titled "Sustaining US Leadership in Space" was released in the Space Foundation's Research and Analysis program. It talks about amending the "Outer Space Treaty", justifying regulations, managing institutional knowledge and increasing the amount of in-house technical work. SSERVI formerly known as the NASA Lunar Science Institute published an article that mentions how a recent report from the Congressional Research Service proves how the US has a supply vulnerability for REMs, many of which have defence applications. The US is serious about supply sustainability and the moon is the only solution, according to that report as well as common sense.

But should the West really be the ones to get their hands on the Moon? Let us not forget that this the civilization that bombs Muslim countries and then equates India with Pakistan for mistreating its "minorities". This is the civilization that pressurises India to not buy oil from Iran because of their political affections to the Saudis. It is these "developed" NATO nations and the US aren't willing to share green tech with India, and yet want us to lower carbon emissions. These are the same people who reprimand us for not taking care of our poor and advise us to disband our space programme and not vie for the stars in order to feed our poor.

Crouching Tiger, Hidden Dragon

The Chinese are already making strides with their Chang'e program. The recent Chang'e 5 mission brought back fusion-ready Helium-3. In the past, China had blocked the export of rare earth elements to Japan. China being the exporter of 80-90 percent of the world's electronics is increasingly becoming more



The Chinese have the most to gain from mining the moon

judicious with export of its resources. All the while other nations worry over where they will procure these limited resources in the future. It is primarily because of China's investments in Africa that they have acquired mineral rights to the most potent of regions in the world. However, even then according to the estimates of Chinese experts, their reserves will not last more than two decades. The moon is the only way out of this and China knows it.

However the problem with the Chinese is that they sport the same supercilious spirit as the West. After the financial meltdown in the West, the Chinese administration decided that this was their time to rise and have been pursuing supremacy over Asia and the rest of the world by hook or by crook. They have annexed a region of Leh/Ladhakh and continue to pressurize India to let go of her sovereign territories including but not limited to Arunachal Pradesh. China has had a long-standing plan to divert the waters of the Indian rivers to boost their own agricultural produce. This could cause the worst famine in human history in the Indian Subcontinent. They have also annexed regions from Japan, Vietnam, Myanmar and continue to claim ownership over what is now known as the South China sea.

The power struggle and wars for resources can get really ugly. There can be no doubt that these countries are aiming for the moon... and literally at that. And they have the means to get there and also tap into the resources. The future does not look very pretty for most people of the world. But this only includes what the major powers plan and are capable of. And not what the other nations are capable of doing. d



The moon has enough rare minerals to meet mankind's energy needs for the next 1000 years